

Script -Controlled-ACL Restrict Record

Access Based on Field value

☒ Introduction

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☒Project Overview *This project focuses on using a script controlled Access Control List (ACL) in ServiceNow to restrict record access based on a specified field value. The ACL script checks the field value and controls whether the user is allowed to access the record. The main purpose of the project is to provide controlled and secure access to ServiceNow records

☒ Milestone-1 Creation of Users and Roles

a)User Creation:

- 1. Log in to the ServiceNow instance with admin access.**
- 2. Create a test user:**
 - **Navigate to User Administration → User**
 - **Click New**
 - **User ID: EEE User**
 - **First name: EEE**

- **Last name: User**
- **Email: eeeuser@gmail.com**
- **Save the record**

The screenshot displays the ServiceNow 'Users' list interface. The browser address bar shows the URL: `dev302430.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253e%253dEEE%26sysparm_...`. The ServiceNow header includes navigation tabs (All, Favorites, History, Workspaces) and a 'Users' button. A search bar is present with the text 'Search'. Below the header, a filter bar shows 'All > Name >= EEE'. The main table lists users with the following columns: User ID, Name, Email, Active, Created, and Updated. The first user is 'EEE User' with email 'eeeuser@gmail.com'. Other users include efren.baucher, elaine.renzi, eldon.sutch, eli.bettner, elisa.gracely, eliseo.wice, ella.pahnke, elmo.dagenais, elmo.gabouer, elvira.blumenthal, and emerv.reek. A pagination bar at the bottom shows '1 to 20 of 460'.

User ID	Name	Email	Active	Created	Updated
EEE User	EEE User	eeeuser@gmail.com	true	2026-09-23 05:36:47	2026-09-23 05:36:47
efren.baucher	Efren Baucher	efren.baucher@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:59
elaine.renzi	Elaine Renzi	elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-09-12 10:30:57
eldon.sutch	Eldon Sutch	eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-09-12 10:31:05
eli.bettner	Eli Bettner	eli.bettner@example.com	true	2012-02-17 19:04:51	2026-09-12 10:31:00
elisa.gracely	Elisa Gracely	elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-09-12 10:30:57
eliseo.wice	Eliseo Wice	eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:58
ella.pahnke	Ella Pahnke	ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-09-12 10:31:03
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-09-12 10:31:04
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:56
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-09-12 10:31:02
emerv.reek	Emerv Reek	emerv.reek@example.com	true	2012-02-17 19:04:49	2026-09-12 10:30:58

b) Roles Creation

- **Navigate to User Administration → Roles**
- **Click New**
- **Name: Enter role name as bb1**
- **Description: Brief role purpose**
- **Click Submit**
- **Create other roles also like bb2,bb3,bb4.**
- **Assign all the roles to the EEE User**

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EEE User | User | ServiceNow

Ask Gemini

dev302430.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D93d85f7047af4f1068a89e20116d43a1%26sysparm_view%3D%26sysparm_record_ta...

YouTubeMapsGmailPrompt Design in A...Learning Course - S...

servicenow

AllFavoritesHistoryWorkspaces

User - EEE User

Search

UpdateSet PasswordDelete

User
EEE User

UpdateSet PasswordDelete

Related Links

[View linked accounts](#)
[View Subscriptions](#)
[Reset a password](#)

Roles (5)Groups (1)DelegatesUser Client Certificates

Role

Search

Actions on selected rows...

Edit...

User = EEE User

Role	State	Inherited	Inheritance Count
bb4	Active	false	
bb1	Active	false	
bb2	Active	false	
bb3	Active	false	
snc_required_script_writer_permission	Active	true	1

1 to 5 of 5

32°C
Partly cloudy

Search

ENG
IN

20:07
24-09-2026

☒ Milestone 2 :Tables Creation

1. In the Application Navigator, search for system Definition → Tables

2. Click New and create a custom table with the following Details.

- Label: Institution Details
- Name: u_institution_details
- Extends for : false

3. Click Submit

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Institution Details
 * Name: u_institution_details
 Application: Global
 Remote Table: ☐

Columns | Controls | Application Access

Table Columns | for text | Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)		40	false
Email	String	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Created by	String	(empty)		40	false

4. Add the following fields to the u_institution_details table:

- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String

5. Save the Table


```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  // Allow only EEE branch users to see EEE records  
  if (gs.hasRole('bb1'))  
  {  
    return true;  
  }  
  // Deny access for all others  
  return false;  
})();
```

8. Click Submit to save the ACL.

9. Impersonate EEE users and open the Student Records list.

10. Copy the table name u_institution_details.list/LIST and paste in the all menu

11. Impersonate Admin and open the Student Records list.

Verify :

- **User with bb1 role can view only EEE branch records**
- **User without the role cannot view any records**
- **Admin user can view all records regardless of branch**

The screenshot shows the ServiceNow interface for configuring an Access Control Rule. The breadcrumb trail is 'Access Control - u_institution_details'. The configuration fields are as follows:

- Type:** record
- Operation:** read
- Application:** Global
- Active:** ☒
- Admin overrides:** ☐
- Advanced:** ☒
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** u_institution_details
- Description:** Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Below the configuration fields is a 'Conditions' section. It contains a blue informational box stating: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access. [More Info](#)'.

Under the 'Conditions' section is a 'Requires role' table with the following data:

Role
bb1

At the bottom, there is a 'Security Attribute Condition' section which is currently empty.

☒ Milestone 4 : Creation of Access Control Lists(Create ACL)

1. Click New to create a new ACL(Create)

2. Enter the following details:

- **Type: record**
- **Operation: create**
- **Name: u_institution_details**
- **Active: true**

3. In the Requires role related list, add:

- **bb2 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify:

- **User with bb1 &bb2 roles can view only EEE branch records**

b New Button

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The top navigation bar includes the ServiceNow logo, 'All', 'Favorites', 'History', and a search bar. The breadcrumb trail is 'Access Control - u_institution_details'. The main form contains the following fields:

- Type: record
- Operation: create
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Advanced: ☐
- Decision Type: Allow If
- Protection policy: -- None --
- Name: u_institution_details
- Description: Allow create for records in u_institution_details, for users with role bb2.

Below the form is a 'Conditions' section with a blue informational box stating: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access. [More Info](#)'

Under the 'Requires role' section, there is a table with one entry:

Role
bb2

The bottom of the image shows a Windows taskbar with the date 24-09-2026 and time 20:35.

☒ Milestone 5 : Creation of Access Control Lists(Write ACL)

1. Click New to create a new ACL(Write)

2. Enter the following details:

- **Type: record**
- **Operation: Write**
- **Name: u_institution_details**
- **Active: true**

3. In the Requires role related list, add:

- **bb3 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify : ● User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The breadcrumb trail is 'Access Control - u_institution_details'. The configuration fields are as follows:

- Type: record
- Operation: write
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Advanced: ☐
- Decision Type: Allow If
- Protection policy: -- None --
- Name: u_institution_details
- Description: Allow write for records in u_institution_details, for users with role bb3.

Below the configuration fields is a 'Conditions' section. It contains a blue informational box stating: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access. [More Info](#)'. Below this box is a 'Requires role' section with a table:

Role
bb3

At the bottom of the 'Requires role' section, there is a 'Security Attribute Condition' section which is currently empty.

☒ Milestone 6 : Creation of Access Control Lists(Delete ACL)

1. Click New to create a new ACL(Delete)

2. Enter the following details:

- **Type: record**
- **Operation: Delete**
- **Name: u_institution_details**
- **Active: true**

3. In the Requires role related list, add:

- **bb4 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify :

● Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.

servicenow

AllFavoritesHistory

Access Control - u_institution_details

Search

u_institution_details

Type

record

Application

Global

Operation

delete

Active

☒

Admin overrides

☒

Advanced

☐

Decision Type

Allow If

Protection policy

-- None --

Name

u_institution_details

Description

Allow delete for records in u_institution_details, for users with role bb4.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role

Role

bb4

Security Attribute Condition

☒ Milestone 7 : Conclusion

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

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dev302430.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

YouTubeMapsGmailPrompt Design in A...Learning Course - S...

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AllFavoritesHistory

Institution Details s

BranchSearch

Actions on selected rows...

New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

1 to 1 of 1

31°C
Mostly cloudy

Search

ENG IN

19:50
23-09-2026